

SALIH DUZEN

Senior Full Stack Software Engineer

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U.S. Permanent Resident (Green Card) — Authorized to work in the U.S. for any employer, no sponsorship required

PROFESSIONAL SUMMARY

Senior Full Stack Software Engineer with 24+ years of hands-on experience architecting and delivering enterprise web applications for government, aviation, healthcare, and commodity trading sectors. Deep expertise in Angular (v8–21), Node.js, TypeScript, REST/Microservices, and relational/NoSQL databases. AI-augmented engineering leader — orchestrates Claude (Anthropic) and GitHub Copilot as everyday pair-programming and code-review partners to accelerate delivery, raise code quality, and onboard new technologies faster. Proven record of delivering mission-critical systems used by millions of citizens (Istanbul Metropolitan Municipality, population 16M+), modernizing legacy platforms, and building secure, scalable applications with OAuth2/OpenID Connect. U.S. Green Card holder based in Maryland, open to on-site, hybrid, or remote roles.

CORE TECHNICAL SKILLS

Frontend: Angular (v8–21), TypeScript, JavaScript (ES6+), RxJS, NgRx, HTML5, CSS3, SASS, Bootstrap, Angular Material, PrimeNG, Progressive Web Apps (PWA)

Backend: Node.js, Express.js, Fastify.js, REST APIs, Microservices, Socket.io (real-time), OAuth2 / OpenID Connect

Databases: PostgreSQL, MySQL, Microsoft SQL Server, Oracle, MongoDB, OrientDB (Graph), SQLite, Redis

DevOps & Tools: Git, Linux (Ubuntu), Nginx, Apache, Docker, CI/CD pipelines, REST integration, Postman

GIS / Geospatial: ESRI ArcGIS integration, geospatial data conversion, map-based public services (large-scale)

AI-Assisted Development: Claude (Anthropic) and GitHub Copilot as daily pair-programming, code-review, and architecture-consultation partners; prompt engineering, AI workflow design, and human-in-the-loop quality control across the full SDLC

Additional: C#.NET, Python (IoT/Raspberry Pi), Survey.js, Power BI / SPSS data export, Agile / Scrum

PROFESSIONAL EXPERIENCE

REDSTAR AVIATION — Senior Frontend Developer 2023 – March 2026

- Architected and delivered an aviation operations management platform on Angular 17 used by dispatch, medical, and air-traffic teams to coordinate air-ambulance missions across multiple regions.
- Integrated Microsoft Identity Server (OAuth2 / OpenID Connect) for enterprise SSO, eliminating shared-credential risk and meeting aviation security standards.
- Built modular Angular components with PrimeNG, Bootstrap, and TypeScript, reducing production defects by an estimated 40% versus the previous codebase through strict typing, lazy loading, and OnPush change detection.
- Implemented integrations with the national pharmaceutical authority and air traffic control feeds, enabling real-time tracking of medical inventory, fuel, passengers, and equipment on board.
- Orchestrated Claude (Anthropic) and GitHub Copilot as daily AI pair-programming assistants — driving code generation, refactoring, security review, and documentation — to ship features faster while maintaining strict aviation-grade quality standards.

HEYTEKNOLOJI — Senior Full Stack Engineer 2022 – 2023

- Led full-stack development of a global B2B cotton exchange platform connecting traders across multiple countries, using Angular 12, Node.js, Express, MongoDB, and OrientDB (graph database for relationship modeling).
- Designed and built REST APIs and real-time data flows with Socket.io, feeding live market data into an AI-based price-forecasting engine used by buyers and sellers.
- Engineered configurable data-export pipelines to Excel, JSON, SPSS, and Power BI, cutting analyst report preparation time from days to minutes.
- Owned end-to-end delivery — frontend architecture, database schema design, REST services, and third-party report integrations.

ALBAYRAK GROUP – PIRIMEDYA — Senior Full Stack Engineer 2021 – 2022

- Designed and built a high-volume web survey platform on Angular 12, Node.js, and Survey.js, used by one of Turkey's largest media groups for audience and market research.

- Improved application performance by 30% through architectural refactoring (lazy modules, query optimization, caching), enabling the team to consistently meet release deadlines.
- Implemented a browser-fingerprint anti-fraud layer to prevent duplicate or manipulated responses, and enabled one-time-per-user surveys via signed reference URLs.
- Built end-to-end data pipelines exporting survey responses to SPSS and Power BI for statistical analysis.

ISTANBUL METROPOLITAN MUNICIPALITY — Senior Software Engineer April 2005 – March 2021

Government technology team serving a metropolitan population of 16+ million citizens.

- Modernized legacy municipal systems by migrating monolithic VB/.NET applications to modern Angular + REST architectures running on Oracle Database, improving service reliability for millions of public-facing requests.
- Led the citywide road-data digitization program: scanned and digitized Istanbul's road network, converted geospatial data into ESRI ArcGIS, and published live road maintenance and repair activity on a public-facing interactive map.
- Designed and delivered a garage automation and fleet maintenance tracking system covering thousands of municipal service vehicles (transit, sanitation, emergency), reducing manual logging and downtime.
- Built infrastructure information systems integrating with GIS layers, RESTful services, and Oracle data warehouses, supporting cross-department coordination.
- Acted as full-stack engineer across the full lifecycle — requirements, architecture, frontend, backend, database, and production support — on multiple concurrent enterprise projects over 16 years.

YARAN ELEKTRONIK / PROCESSING WORLD — Software Developer 2003 – 2005

- Built C#.NET Windows Forms applications and MySQL-backed personnel attendance systems, including serial-port integration with biometric (fingerprint) devices and automated transfer of attendance data into accounting systems.

SELECTED PROJECTS

- **D3M.io** — Multi-category classifieds platform (vehicles, real estate, marketplace, jobs, services, rentals, community) with auth, listings, messaging, favorites, and media. Built end-to-end on **Angular 21, Node.js, Fastify, PostgreSQL, and REST APIs**.
- **Sevkiyatvar.com** — Load-matching marketplace connecting freight transporters and customers across Turkey. Built end-to-end on Angular, Node.js, OrientDB, and REST APIs. Mobile experience delivered as an installable Progressive Web App (PWA).
- **Smartdecibel.com** — IoT noise-monitoring platform combining a Raspberry Pi sensor (Python) with an Angular 12 / Node.js dashboard. Captures ambient decibel levels, syncs via REST, and stores readings offline in SQLite during connectivity loss, triggering threshold-based alerts.
- **Istinye State Hospital Food Hall Control System** — Designed and delivered a turnstile/access-control integration for a state hospital cafeteria, using C#.NET and MySQL with serial-port communication and automated daily reporting.

EDUCATION

Bachelor of Science, Software Engineering — Istanbul Aydin University, Istanbul, Turkey

Associate Degree, Computer Programming — Cukurova University, Adana, Turkey

LANGUAGES

English (Working Proficiency) | Turkish (Native)